

“ **TRUTH IN MEDICINE IS AN UNATTAINABLE GOAL, AND THE ART AS DESCRIBED IN BOOKS IS FAR BENEATH THE **KNOWLEDGE** OF AN **EXPERIENCED** AND THOUGHTFUL **PHYSICIAN.** ”**

Al-Razi, Arab physician, 10th century



Arab doctor and chemist al-Razi's belief in practical experimentation on substances led him to propose an early classification of elements.

MUHAMMAD IBN ZAKARIYA

AL-RAZI (Rhazes) was one of the greatest physicians of the Arab world. Around 900, he wrote *Al-Shukuk ala Jalinus* (*Doubts About Galen*), in which he **criticized Galen's theory of the four humors** (see 75–250). He rejected the notion that a balance of these humors was necessary for the health of the patient, and



AL-RAZI (c.865–925)

Born in Rayy, Mesopotamia (now in Iran), al-Razi was a physician and philosopher, as well as an alchemist. He encouraged experimentation as a means of discovery and his clinical notes became a key medieval medical text. He headed a hospital in Rayy, and then two in Baghdad. Among his innovations was the first recorded clinical trial—on patients with meningitis.

dismissed the idea that body temperature is automatically raised or lowered when a patient drinks warm or cold fluids. His clinical practices were advanced for this time; he ran a psychiatric ward, and he wrote a treatise attacking untrained physicians. The *Kitab al-Hawi* (*Comprehensive Book*), a collection of his clinical notes, ran to 23 volumes, and contained **medical diagnoses**, including the first description of **hay fever** (or rose-cold). He also wrote a monograph, *Kitab al-Judwar wal Hasba* (*Treatise on the Smallpox and Measles*), which was the first work to detail the symptoms of smallpox, although his explanation—that the disease was caused by the impurities from menstrual blood that stay in the fetus during pregnancy and then bubble up to the skin in later life—betrayed a belief in sympathetic magic. He was particularly concerned with **preventing blindness caused by smallpox pustules**, and advocated regularly bathing the eyes in rose-water.

An alchemist as well as a physician, al-Razi devised a **classification of elements** into spirits and metals and minerals. He divided the latter into stones, vitriols, boraces, salts, and other substances, and gave a detailed account of the behavior of each under various processes, such as

melting and extraction. He described the **distillation of kerosene and petroleum** from crude oil and gave recipes for preparing **hydrochloric and sulfuric acids**.

Around 920, Arab astronomer and mathematician **al-Battani** (c.858–929) proffered greater insights into the working of the **planispheric astrolabe**—a device with a number of

overlapping plates for making astronomical observations. Although al-Fazari was the first to describe it in the 8th century, al-Battani worked out the

mathematics underlying the instrument. He presented formulas in spherical trigonometry, replacing Ptolemy's geometrical methods.

map of bodies on the celestial sphere

star pointer indicates position of specific star

mater, or main section into which latitude plates slot

rotating bar

ecliptic ring shows path of Sun through sky

Astrolabe

The user of an astrolabe adjusted its moveable parts to indicate a specific date or time, and the markings on the plates would then indicate the position of the various heavenly bodies.



c.900 Islamic mathematician Abu Kamil develops al-Khwarizmi's algebra, dealing with powers greater than 2

c.912 Qusta ibn Luqa writes a treatise on numbness

900–30 Al-Razi describes symptoms of smallpox

900–30 Al-Razi criticizes Galenic humor theory

c.920 Al-Battani discovers mathematical principles underlying the astrolabe

c.925 Al-Farabi writes treatise on music therapy

c.927 Earliest dated surviving astrolabe is built by Islamic astronomer Nastulus